Preconditions

`*this` has an associated asynchronous result.

Effects

Stores `r` in the asynchronous result associated with `*this` if `ResultType` isn't `void`. Marks the asynchronous result as having a stored value. Schedules the associated asynchronous result to be made *ready* when the current thread exits.

Postconditions

The asynchronous result associated with `*this` has a stored value but isn't *ready* until the current thread exits. Threads blocked waiting for the asynchronous result will be unblocked when the current thread exits.

Throws

An exception of type `std::future_error` with an error code of `std::future_errc::promise_already_satisfied` if the asynchronous result already has a stored value or exception. Any exceptions thrown by the copy-constructor or move-constructor of `r`.

Synchronization

Multiple concurrent calls to `set_value(), set_value_at_thread_exit(), set_exception()`, and `set_exception_at_thread_exit()` are serialized. The completion of the thread that made a successful call to `set_value_at_thread_exit()` happens-before a call to `std::future<ResultType>::get()` or `std::shared_future<ResultType>::get()`, which retrieves the exception stored.

Std::Promise::Set_Exception Member Function